USER INTERFACE DESIGN

EXPERIMENT 3

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Command Line Interface

```
import random


options = ["rock", "paper", "scissors"]
system_choice = random.choice(options)


print("□ Rock Paper Scissors - CLI Version")
print("Choices: rock / paper / scissors")


user_choice = input("Enter your choice: ").lower()


  ifuser_choice not in options:
     print("✕Invalid choice")
else:
   print(f"Computer chose: {system_choice}")
```

```
ifuser_choice == system_choice:
    print("□ It's a tie!")
  elif(user_choice == "rock" and system_choice == "scissors") or \
    (user_choice == "paper" and system_choice == "rock") or \
    (user_choice == "scissors" and system_choice == "paper"):
    print("□ You win!")
else:
    print("□ You lose!")
```

OUTPUT:

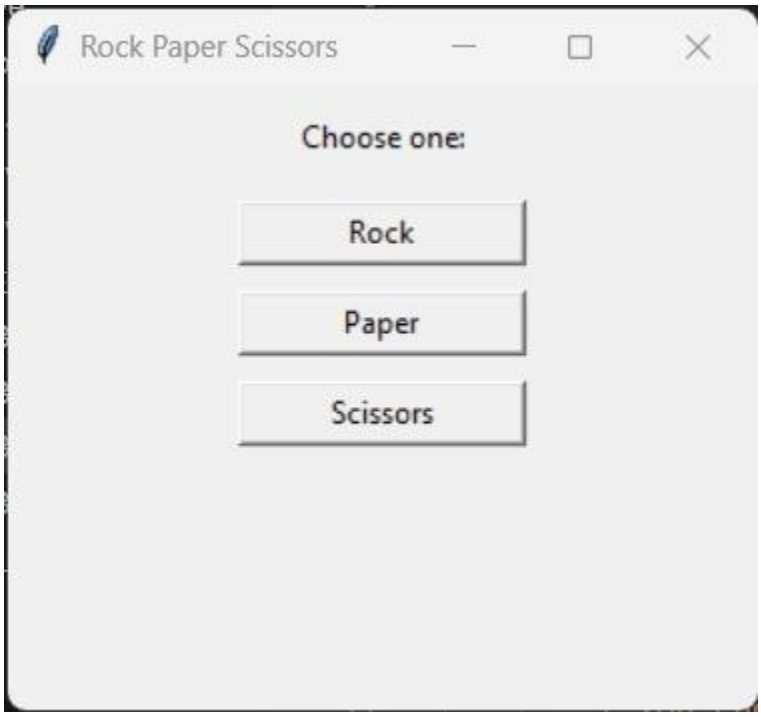
```
PS C:\Users\bivis\Downloads\UID_EXP3> & C:/Users/bivis/AppD
3/exp3_cli.py
🎮 Rock Paper Scissors - CLI Version
Choices: rock / paper / scissors
Enter your choice: rock
Computer chose: scissors
🎉 You win!
PS C:\Users\bivis\Downloads\UID_EXP3>
```

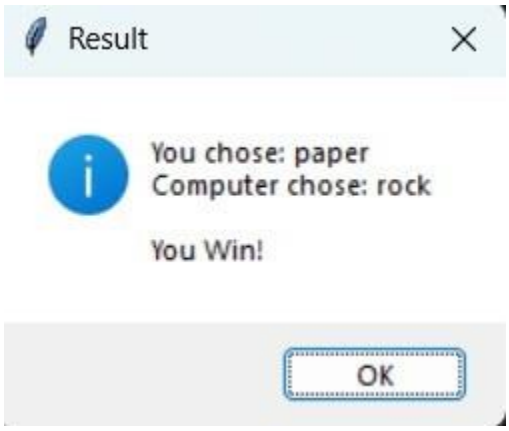
1. Graphical User Interface

```
import tkinter as tk
import random
from tkinter import messagebox
```

```
choices = ["rock", "paper", "scissors"]

defplay_game(user_pick):
    computer_pick = random.choice(choices)

    ifuser_pick == computer_pick:
        result = "It's a Tie!"
    elif(user_pick == "rock" and computer_pick == "scissors") or \
        (user_pick == "paper" and computer_pick == "rock") or \
        (user_pick == "scissors" and computer_pick == "paper"):
        result = "You Win!"
    else:
        result = "You Lose!"

    messagebox.showinfo(
        "Result",
        f"You chose: {user_pick}\nComputer chose:
{computer_pick}\n\n{result}"
    )

window = tk.Tk()
window.title("Rock Paper Scissors")
window.geometry("300x250")

tk.Label(window, text="Choose one:").pack(pady=10)

tk.Button(window, text="Rock", width=15, command=lambda:
play_game("rock")).pack(pady=5)
tk.Button(window, text="Paper", width=15, command=lambda:
play_game("paper")).pack(pady=5)
tk.Button(window, text="Scissors", width=15, command=lambda:
play_game("scissors")).pack(pady=5)

window.mainloop()
```

OUTPUT:





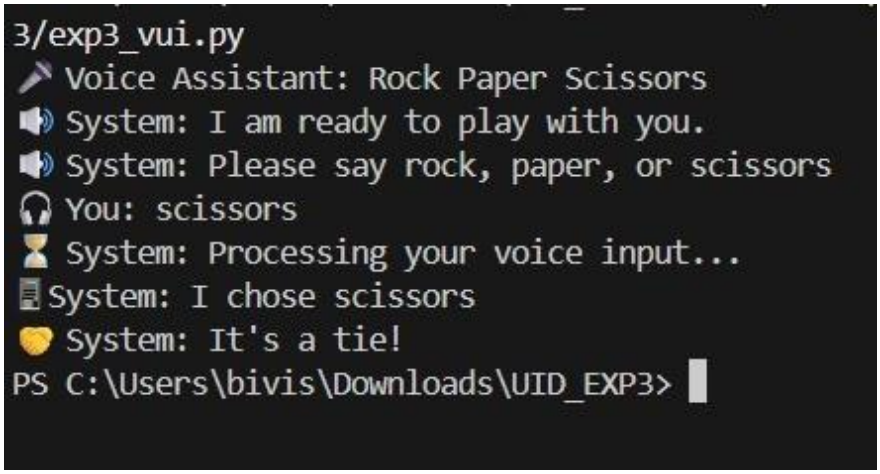
2. Voice User Interface

import random

```
import time

choices = ["rock", "paper", "scissors"]
computer = random.choice(choices)

print("▯ Voice Assistant: Rock Paper Scissors")
time.sleep(0.5)
print("□ Please say rock, paper, or scissors")

whileTrue:
    user= input(" ▯  You: ").lower()

    ifuser not in choices:
        print(" ⚠ System: Please say a valid option")
        continue

    print("□ System is thinking...")
    time.sleep(0.5)
```

```
    print(f"    System chose: {computer}")


    ifuser== computer:
        print("  System:  It's  a  tie!")
      elif(user == "rock" and computer == "scissors") or \
        (user == "paper" and computer == "rock") or \
        (user == "scissors" and computer == "paper"):
        print("  System:  You  win!")
    else:
        print("  System: You lose!")
    break
```

OUTPUT:



3. User Satisfaction Comparison

```
defsurvey():
    try:
        print("Rate your satisfaction with the following interfaces (1–5):")

        cli_satisfaction  =  int(input("CLI  (1-5):  "))
        gui_satisfaction  =  int(input("GUI  (1-5):  "))
        vui_satisfaction = int(input("VUI (1-5): "))

        print("\nYour  satisfaction  ratings:")
        print(f"CLI:        {cli_satisfaction}")
        print(f"GUI:       {gui_satisfaction}")
        print(f"VUI: {vui_satisfaction}")

        #Calculate the average satisfaction
        avg_satisfaction = (cli_satisfaction + gui_satisfaction +
vui_satisfaction) / 3
        print(f"\nAverage Satisfaction Score: {avg_satisfaction:.2f}")

    except ValueError:
        print("Invalid input! Please enter numbers between 1 and 5.")


#Runthe survey function
if__name__== "__main__":
    survey()
```

OUTPUT:

```
3/exp3_compare.py
Rate your satisfaction with the following interfaces (1-5):
CLI (1-5): 4
GUI (1-5): 5
VUI (1-5): 4

Your satisfaction ratings:
CLI: 4
GUI: 5
VUI: 4

Average Satisfaction Score: 4.33
PS C:\Users\bivis\Downloads\UID_EXP3>
```